A Comparative Text Mining Analysis of AI and Non-AI Research Papers Using TF-IDF and Unsupervised Learning

Jahid Mahmud Turag
ID: 21-45638-3
American International University–Bangladesh (AIUB)

Md. Fahim Shahriare Sopnil
ID: 22-46523-1
American International University–Bangladesh (AIUB)

Atikur Rahman Khan Akash
ID: 22-46156-1
American International University–Bangladesh (AIUB)

Md. Istiak Zaman Prapto
ID: 21-45643-3
American International University–Bangladesh (AIUB)

***Abstract*—This study performs a comparative text mining analysis of research papers in Artificial Intelligence/Machine Learning (AI/ML) and non-AI domains using TF-IDF-based feature extraction and unsupervised learning. Two separate corpora were preprocessed and transformed into TF-IDF vectors to identify distinctive terms and contrasting patterns. K-means clustering and hierarchical clustering (cosine distance) were applied to explore structural variation across documents. Cluster quality was assessed using the Elbow criterion, Silhouette score, and Adjusted Rand Index (ARI) by comparing cluster assignments with known labels. Principal Component Analysis (PCA) was used for visualization of separation in a lower-dimensional space. Results show clear contrast: AI/ML documents are dominated by model-, algorithm-, and data-related terminology, while non-AI documents emphasize domain-specific and analysis-oriented terms. Overall, the experiment demonstrates that TF-IDF with unsupervised clustering can meaningfully separate AI-related research from broader non-AI literature.**

***Index Terms*—Text Mining, TF-IDF, Artificial Intelligence, Unsupervised Learning, K-means Clustering, Hierarchical Clustering, PCA**

I. Introduction

This work compares AI/ML research papers with non-AI research papers using a text mining pipeline. After cleaning and normalization, documents are converted into TF-IDF representations, then clustered using unsupervised learning methods to quantify separability between the two corpora. The intent is to show that distinctive vocabulary patterns (e.g., model/algorithm/data emphasis in AI/ML) can be detected and leveraged for automatic categorization.

II. Methodology

Fig. 1 summarizes the end-to-end workflow used in this study. The details of the pipeline are given below:

A. Data Collection (Web Scraping)

Text was collected from online sources via web scraping and structured for analysis.

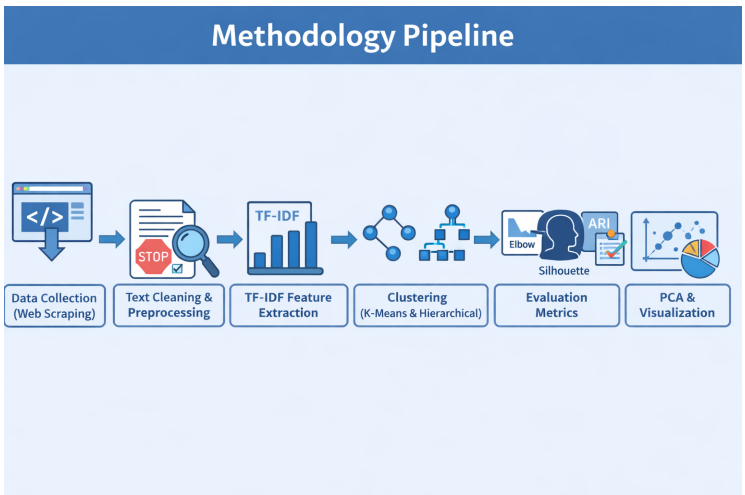

Fig. 1. Methodology pipeline for comparing AI/ML and non-AI research papers using TF-IDF and unsupervised learning.

B. Text Cleaning and Preprocessing

Raw text was processed through standard steps including tokenization, stopword removal, punctuation handling, and normalization (e.g., lemmatization) to reduce noise and improve comparability across documents.

C. TF-IDF Feature Extraction

Term Frequency–Inverse Document Frequency (TF-IDF) was used to convert documents into numeric vectors, emphasizing discriminative terms and reducing the impact of common, non-informative words [1].

D. Clustering Techniques

1) K-means: K-means was applied to partition documents by minimizing within-cluster variance in TF-IDF space.

2) Hierarchical Clustering: Hierarchical clustering using cosine distance was used to build a dendrogram structure, providing an interpretable view of document similarity.

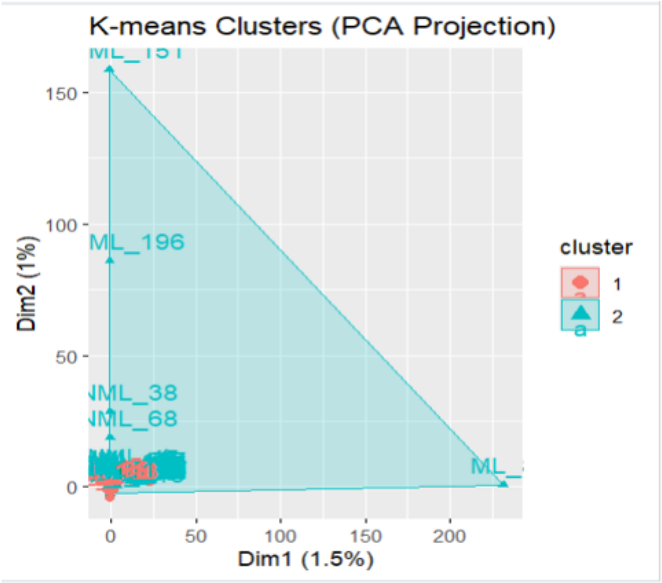

Fig. 2. K-means clustering results visualized using a 2D PCA projection (Dim1 vs. Dim2).

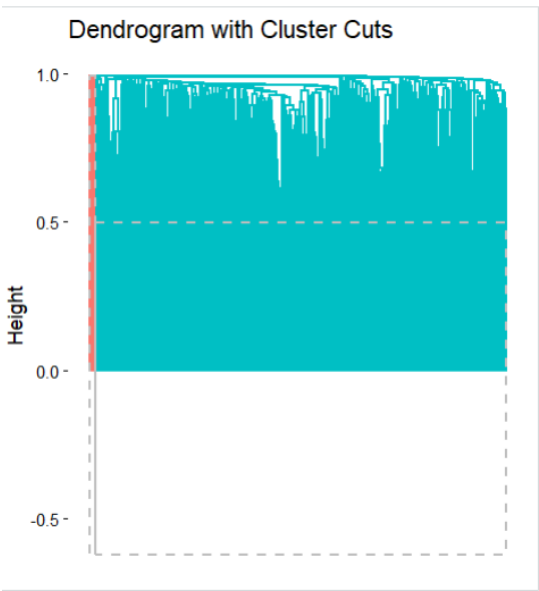

Fig. 3. Hierarchical clustering dendrogram with the selected cut level used to form clusters.

E. Evaluation and Visualization

Clustering quality was assessed using:

- Elbow criterion to guide selection of the number of clusters,
- Silhouette score for cohesion/separation assessment [2],
- Adjusted Rand Index (ARI) against known labels [3].

PCA was used for 2D visualization of separation.

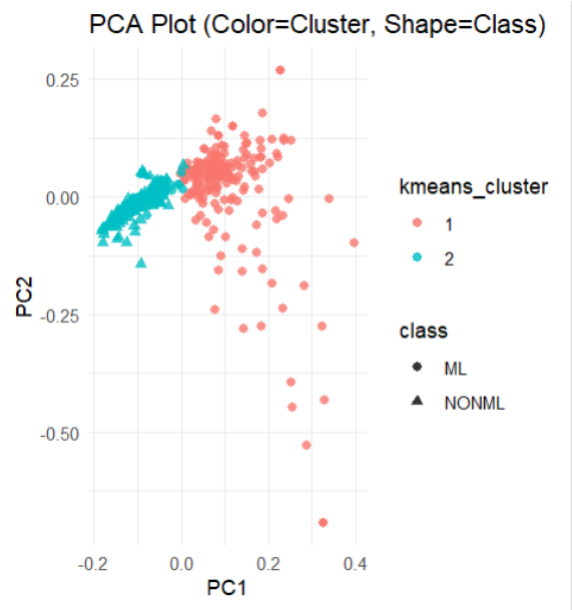

Fig. 4. PCA visualization of TF-IDF document vectors: color indicates K-means cluster assignment and shape indicates the corpus class (ML vs. NONML).

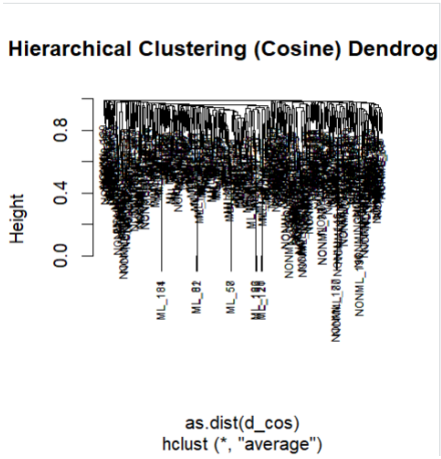

Fig. 5. Hierarchical clustering dendrogram using cosine distance and average linkage on TF-IDF document vectors.

III. Implementation

A. Software and Libraries

Analysis was conducted in **R**, using typical text-mining and visualization workflows (e.g., tokenization, TF-IDF construction, clustering, and plotting).

B. Pipeline Steps

1) Web scraping of target sources into a structured dataset.
2) Text preprocessing (tokenization, stemming/lemmatization, stopword removal).
3) TF-IDF vectorization.
4) K-means and hierarchical clustering in TF-IDF space.
5) Validation via Elbow/Silhouette/ARI.
6) PCA visualization of document separation.

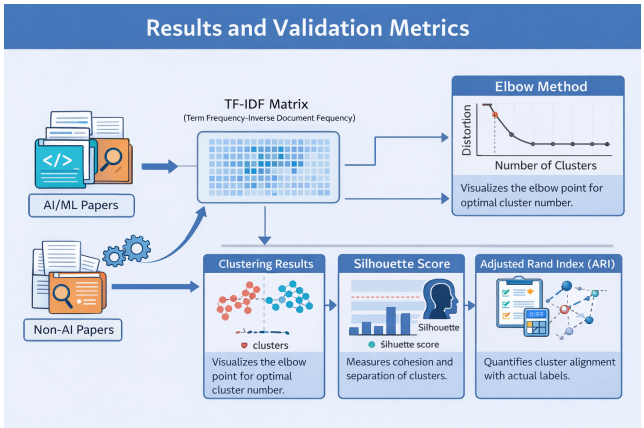

Fig. 6. Summary of clustering validation: TF-IDF representation, clustering outcomes, and evaluation using Elbow, Silhouette, and ARI metrics.

C. Challenges

Noisy text, high dimensionality, and cluster-count selection were addressed using preprocessing, TF-IDF normalization, and validation with Elbow and Silhouette methods.

IV. Results and Analysis

Fig. 6 summarizes the evaluation workflow and key validation outputs used to assess clustering quality.

TF-IDF highlighted discriminative terms that characterize the learned groupings. Documents in the AI/ML corpus were dominated by model-, algorithm-, and data-centric terminology, whereas the non-AI corpus emphasized domain-specific and analysis-oriented language. K-means clustering produced two principal groups that were broadly consistent with the AI/ML versus non-AI split. Fig. 6 provides a visual summary of the evaluation workflow, while the quantitative outcomes are consolidated in Table I. The Elbow criterion supported the selected number of clusters, and the Silhouette score indicated meaningful within-cluster cohesion and between-cluster separation in TF-IDF space. In addition, hierarchical clustering using cosine distance with average linkage yielded dendrogram structures that were consistent with the K-means partitioning. To measure agreement between unsupervised assignments and the known corpus labels, the Adjusted Rand Index (ARI) was computed (Table I). Finally, PCA-based scatter plots were used to visualize separability and cluster structure in a lower-dimensional projection, complementing the metric-based analysis.

V. Discussion

Overall, the results indicate that a TF-IDF representation coupled with unsupervised learning can reliably capture topical differences between AI/ML and non-AI research papers. The term-weight patterns suggest that AI/ML documents share a relatively consistent vocabulary centered on models, algorithms, and data, while non-AI documents exhibit more domain-specific language and broader analytical phrasing. This difference likely explains why two dominant clusters emerged and why the selected k was supported by the Elbow criterion and the Silhouette score (Table I). The hierarchical dendrogram further reinforces this structure by revealing stable groupings under cosine-based similarity, which is well-suited for sparse TF-IDF vectors.

TABLE I
Clustering evaluation summary (placeholders with the K-value (outcome).

Method / Metric	Measure	Value	Interpretation
Elbow (K-means)	Within-cluster SSE trend	$k = [2]$	Elbow indicates diminishing returns beyond k.
Silhouette	Cohesion vs. separation	*[0.xx]*	Higher is better; near 1 implies strong separation.
ARI (vs. labels)	Agreement with ML/NONML	*[0.xx]*	0 random; 1 perfect label alignment.
Hierarchical (cosine, avg.)	Similarity tree structure	—	Dendrogram supports cluster structure seen in K-means.

At the same time, partial overlap in the PCA space indicates that the boundary is not perfectly separable: interdisciplinary papers and general methodological terms can blur cluster assignments, reducing alignment with labels (as reflected by ARI). Because PCA is a low-dimensional projection, apparent overlap may also reflect information loss rather than true similarity. Future work can strengthen robustness by expanding the corpus, controlling for document length and source bias, and comparing TF-IDF with alternative representations (e.g., topic models or transformer embeddings) while reporting stability across multiple random initializations and resamples.

VI. Conclusion

This study shows that TF-IDF with unsupervised clustering can effectively distinguish AI/ML research papers from non-AI research papers based on term usage patterns. The approach can support automatic categorization and exploratory analysis across large document collections.

References

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